Streamlit App Deployment

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- Build and deploy a machine learning application locally
- Deploy ML application app on Web

What is Streamlit?

From the data science pipeline, one of the most important steps is ***model deployment***.

We have a lot of options in python for deploying our model. Some popular frameworks are ***Flask*** and ***Django***.

But the issue with using these frameworks is that we should have some knowledge of ***HTML***, ***CSS***, and ***JavaScript***.

Using streamlit you can ***deploy*** any ***machine learning model*** and any ***python project*** with ease and without worrying about the ***frontend***.

Streamlit is very ***user-friendly***.

Why Streamlit?

Streamlit is a free and open-source framework to rapidly build and share machine learning and data science web apps.

It is a Python-based library specifically designed for machine learning engineers.

It is compatible with the majority of Python libraries (e.g. pandas, matplotlib, seaborn, plotly, Keras, PyTorch, SymPy(latex)).

Less code is needed to create amazing web apps.

Data caching simplifies and speeds up computation pipelines.

Python Virtual Environment

A virtual environment is an isolated environment for python projects.

It allows you to create an isolated environment for each python project.

This makes it easier for us to manage packages and dependencies throughout projects, especially where they share the same dependencies.

Installing Streamlit with Promit

Windows Command Promit

```
cd C:\Users\xxx\Documents\stremlit_projects

python -m venv streamlitenv2

streamlitenv2\Scripts\activate
streamlitenv2\Scripts\Activate.ps1

(streamlitenv1) C:\Users\xxx>

pip install streamlit

streamlit hello
python -m streamlit hello


streamlit run app.py
python -m streamlit run app.py
```

Windows Power Shell

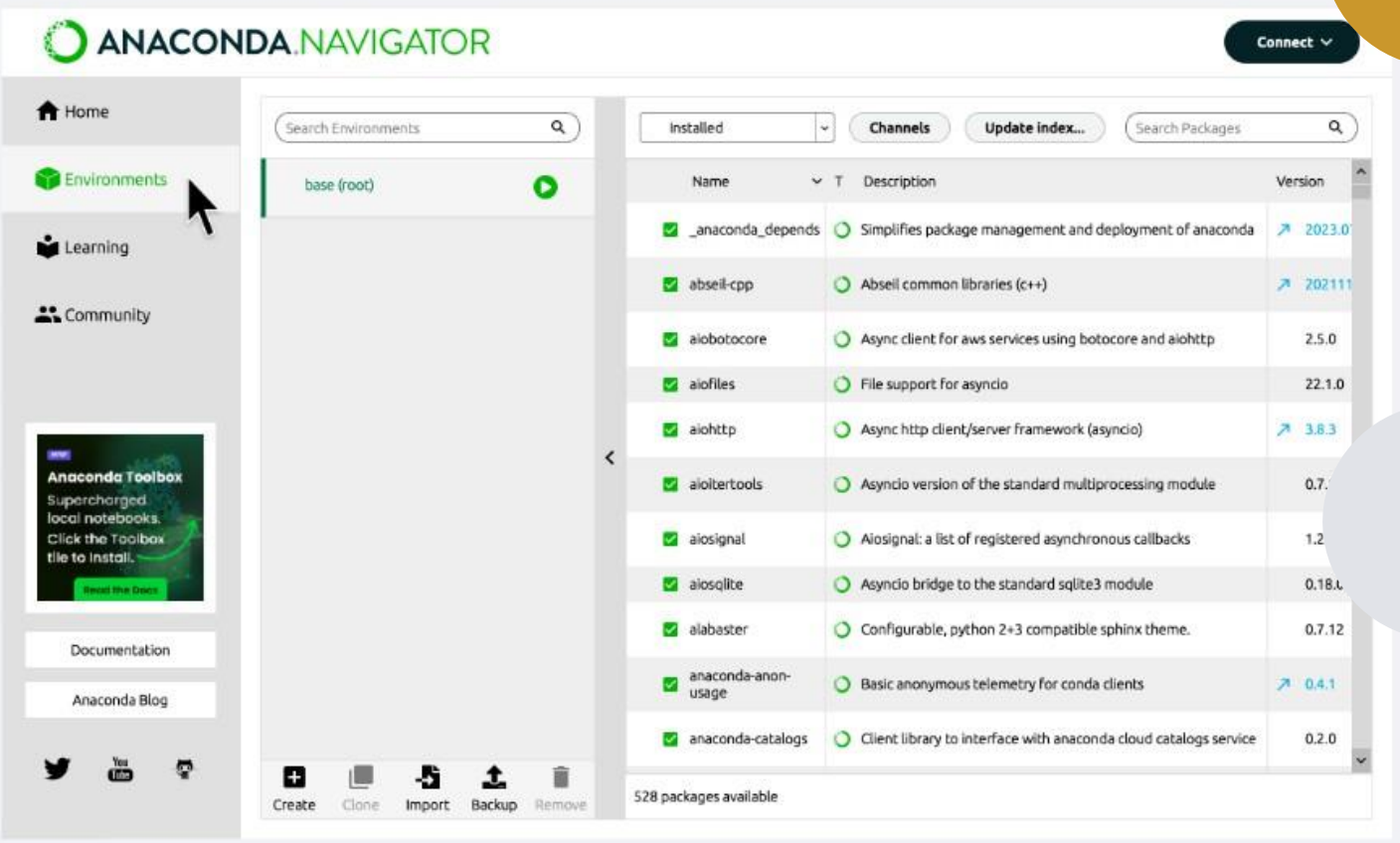
```
cd C:\Users\xxx\Documents\stremlit_projects

python -m venv streamlitenv2


streamlitenv2\Scripts\activate
streamlitenv2\Scripts\Activate.ps1

(streamlitenv1) C:\Users\xxx>

pip install streamlit

streamlit hello
python -m streamlit hello


streamlit run app.py
python -m streamlit run app.py
```

Installing Streamlit: Linux: Ubuntu

```
conda create -n streamlit_env1

conda activate streamlit_env1

pip install streamlit

streamlit hello
python -m streamlit hello

streamlit run app.py
python -m streamlit run app.py
```

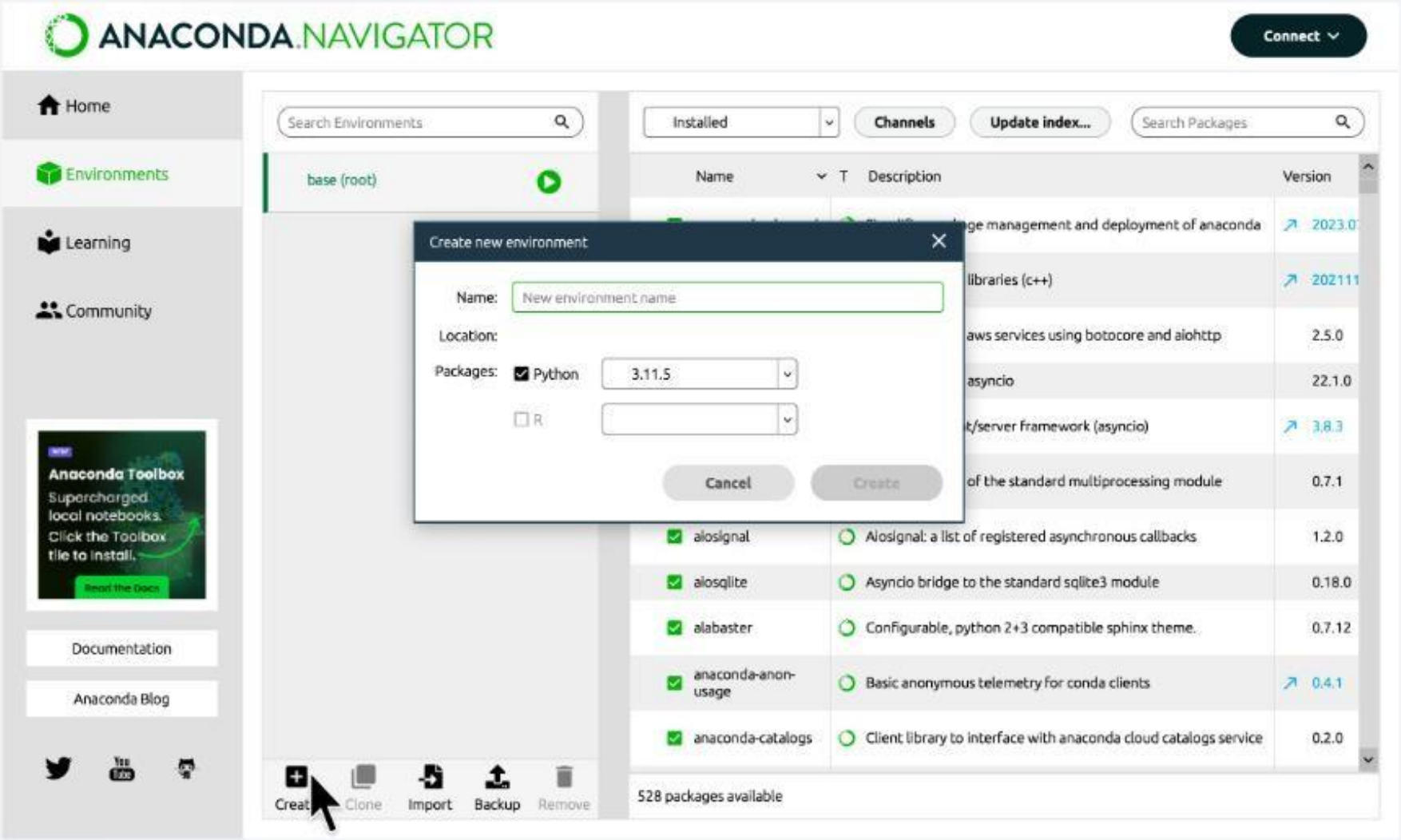
Installing Streamlit using Anaconda

Create	Create an environment using Anaconda Navigator
Open	Open Anaconda Navigator (the graphical interface included with Anaconda Distribution).
Click	In the left menu, click "Environments".



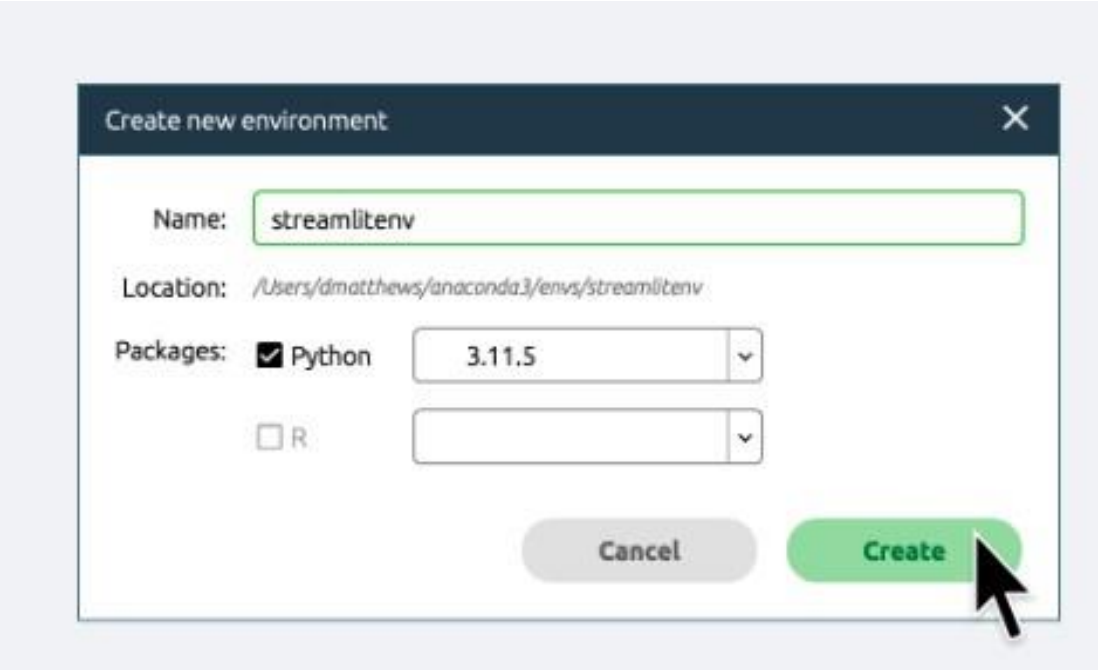
Installing Streamlit using Anaconda

- At the bottom of your environments list, click "**Create**".



Installing Streamlit using Anaconda

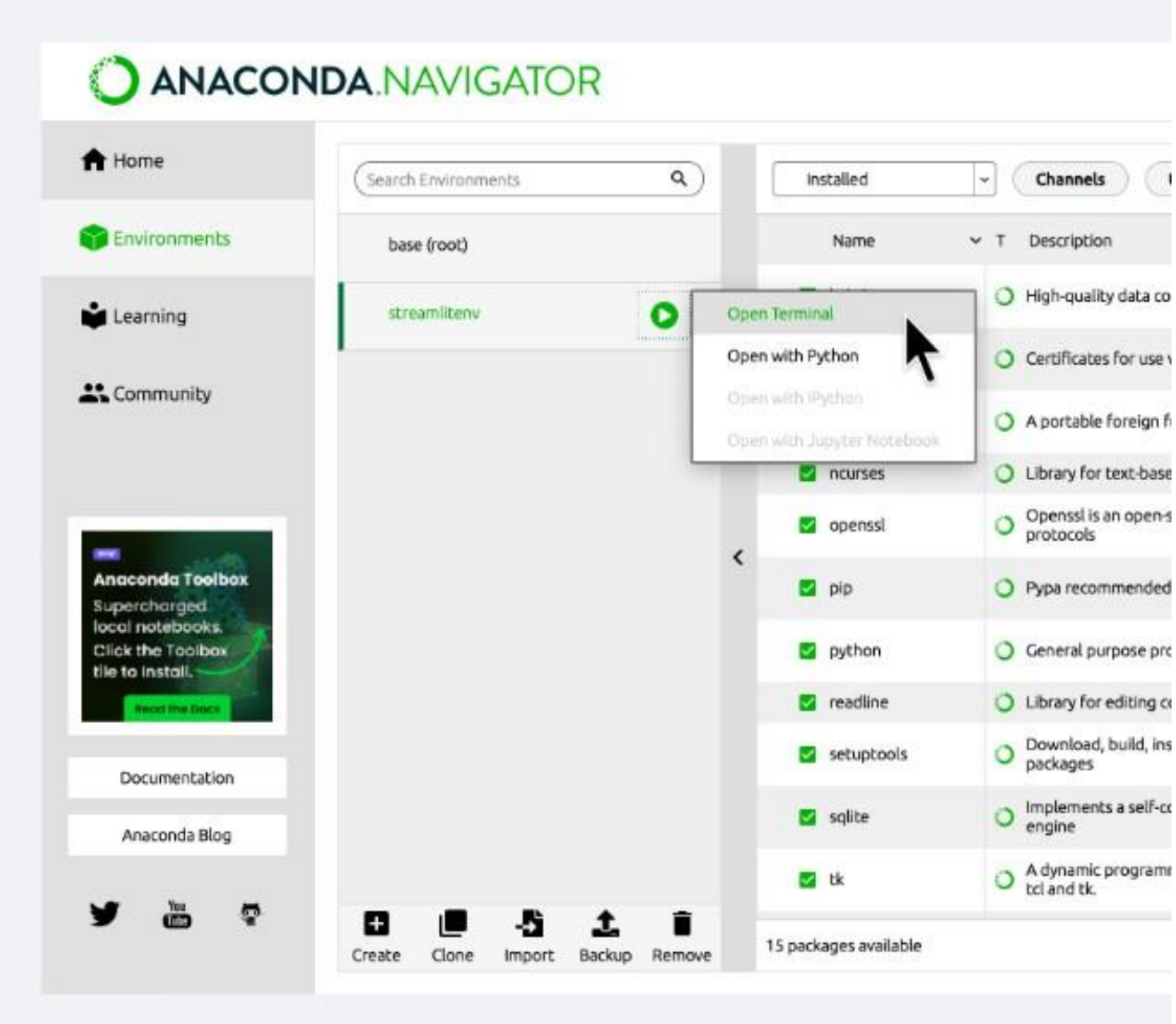
- Enter "streamlitenv" for the name of your environment.
- Click "**Create**."



Installing Streamlit using Anaconda

- Click the green play icon (play_circle) next to your environment.

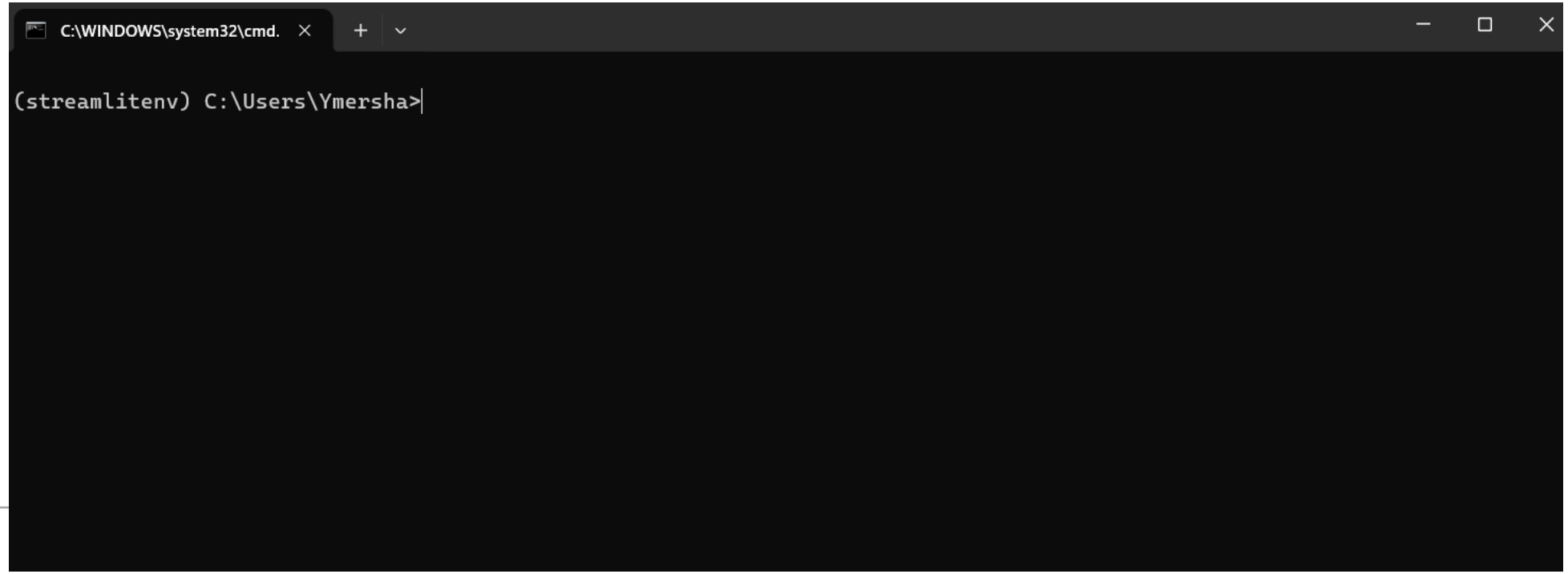
- Click "Open Terminal."



Installing Streamlit using Anaconda

- A terminal will open with your environment activated.

- Your environment's name will appear in parentheses at the beginning of your terminal's prompt to show that it's activated.



Installing Streamlit using Anaconda

- In your terminal, type:

```
pip install streamlit

streamlit hello
python -m streamlit hello
```

- The Streamlit Hello example app will automatically open in your browser.

- If it doesn't, open your browser and go to the localhost address indicated in your terminal, typically http://localhost:8501.

Streamlit: Hello World

```
import streamlit as st

st.write("Hello World")
```

- Download VS Code & install it: https://code.visualstudio.com/download
- Create project folder
- Open **VS Code** Editor with the new project
- Create a Python file named ***app.py*** in your project folder
- Copy the following code into app.py and save it.
- Activate the streamlit environment
- In your terminal, type: ***streamlit run app.py***
- If this doesn't work, use the long-form command: ***python -m streamlit run app.py***
- When you're done, you can stop your app with Ctrl+C in your terminal or just by closing your terminal

Display texts with Streamlit

st.title(): This function allows you to add the title of the app.

st.header(): This function is used to set header of a section.

st.markdown(): This function is used to set a markdown of a section.

st.subheader(): This function is used to set sub-header of a section.

st.caption(): This function is used to write caption.

st.code(): This function is used to set a code.

st.latex(): This function is used to display mathematical expressions formatted as LaTeX.

Markdown

```
st.markdown("# This is a markdown")

st.markdown("## This is a markdown")

st.markdown("### This is a markdown")

st.markdown("#### This is a markdown")

st.markdown("##### This is a markdown")

st.markdown("###### This is a markdown")
```

Title, Header, Subheader, text, & write

```
st.title("This is a title")

st.header("This is a header")

st.subheader("This is a subheader")

st.text("This is a text")

st.write("This is a write")
```

Write Function

- Using write function, we can also display code in ***coding format***.

- This is not possible using st.text(" ").

```
# Write text
st.write("Text with write")

# Writing python inbuilt function range()
st.write(range(10))

st.text(range(10))
```

Success, Info, Warning, Error, & Exception

```
# Success
st.success("Success")

# Information
st.info("Information")

# Warning
st.warning("Warning")

# Error
st.error("Error")

# Exception - This has been added later
exp = ZeroDivisionError("Trying to divide by Zero")
st.exception(exp)
```

Caption, Code, & latex

- st.caption("this is the caption")
- st.code("x=2021")
- st.latex(r''' a+a r^1+a r^2+a r^3 ''')

Display Images

- Using write function, we can also display code in ***coding format***.

- This is not possible using st.text(" ").

```
# import Image from pillow to open images
from PIL import Image

img = Image.open("streamlit.png")

# display image using streamlit
st.image(img, width=200, caption='streamlit logo')
```

Display an image, video or audio file with Streamlit

- st.image("kid.jpg")
- st.audio("Audio.mp3")
- st.video("video.mp4")

Input widgets 1

- Streamlit has various widgets that allow you to interact directly with your app.
- Widgets are the most important user interface components.
- ***st.checkbox()***: This function returns a Boolean value. When the box is checked, it returns a True value, otherwise a False value.
- ***st.button()***: This function is used to display a button widget.
- ***st.radio()***: This function is used to display a radio button widget.
- ***st.selectbox()***: This function is used to display a select widget.
- ***st.multiselect()***: This function is used to display a multiselect widget.
- ***st.select_slider()***: This function is used to display a select slider widget.

- ***st.slider()***: This function is used to display a slider widget.

Checkbox

- A checkbox returns a **boolean value**.
- When the box is checked, it returns a **True** value else returns a **False** value.

```
# title of the checkbox is 'Show/Hide'
# display the text if the checkbox returns True value

if st.checkbox("Show/Hide"):
    st.text("Showing the widget")
```

Button

- **st.button()** returns a **boolean value**.
- It returns a **True** value when clicked else returns **False**.

```
# Create a simple button that does nothing
st.button("Click me for no reason")

# Create a button, that when clicked, shows a text
if(st.button("About")):
    st.text("This button is clicked")
```

Radio Button

```
status = st.radio("Select Gender: ", ('Male', 'Female'))

if (status == 'Male'):
    st.success("Male")
else:
    st.success("Female")
```

```
status1 = st.radio("Select Gender: ", ('xxx', 'yyy', 'zzz'))

st.write("Your status is: ", status1)
```

Selection Box & Multi-Selectbox

```
hobby = st.selectbox("Hobbies: ", ['Dancing', 'Reading', 'Sports'])

st.write("Your hobby is: ", hobby)
```

```
hobbies = st.multiselect("Hobbies: ",  ['Dancing', 'Reading', 'Sports'])

st.write("You selected", len(hobbies), 'hobbies')
```

Slider & Select_Slider

```
level = st.slider("Select the level", 1, 5)

# .format() is used to print value of a variable at a
specific position

st.text('Selected: {}'.format(level))
```

```
st.select_slider('Pick a mark', ['Bad', 'Good', 'Excellent'])
```

Input widgets 2

- ***st.number_input()***: This function is used to display a numeric input widget.
- ***st.text_input()***: This function is used to display a text input widget.
- ***st.date_input()***: This function is used to display a date input widget to choose a date.
- ***st.time_input()***: This function is used to display a time input widget to choose a time.
- ***st.text_area()***: This function is used to display a text input widget with more than a line text.
- ***st.file_uploader()***: This function is used to display a file uploader widget.
- ***st.color_picker()***: This function is used to display color picker widget to choose a color.

Input widgets 2

```
st.number_input('Pick a number', 0,10)

st.text_input('Email address')

st.date_input('Travelling date')

st.time_input('School time')

st.text_area('Description')

st.file_uploader('Upload a photo')

st.color_picker('Choose your favorite color')
```

Progress and status bar with Streamlit

- ***st.balloons()***: This function is used to display balloons for celebration.
- ***st.progress()***: This function is used to display a progressbar.
- ***st.spinner()***: This function is used to display a temporary waiting message during execution.

```
st.balloons()

st.subheader("progress bar")
st.progress(10)

import time
st.subheader("with the execution")
with st.spinner('Wait for it...'):
    time.sleep(10)
```

Sidebar and Container

- Sidebar and container are used to organize your app on your page .

st.sidebar() will make element pinned to the left, allowing users to focus on the content in your app.

```
st.sidebar.title("This is side bar ")
st.sidebar.button("Click ")
st.sidebar.radio("Pick Gender", ["Male", "Feamle"] )
```

st.container() is used to create an invisible container where you can put elements in order to create a useful arrangement and hierarchy.

```
container = st.container()
container.write("This is written inside the container")
st.write("This is written inside the container")
```

Display graphs with Streamlit: pyplot

```
import matplotlib.pyplot as plt
import numpy as np

rand=np.random.normal(1, 2, size=20)

fig, ax = plt.subplots()

ax.hist(rand, bins=15)

st.pyplot(fig)
```

random.normal(): generates an array of 20 random numbers with a mean of 1 & a standard deviation of 2.

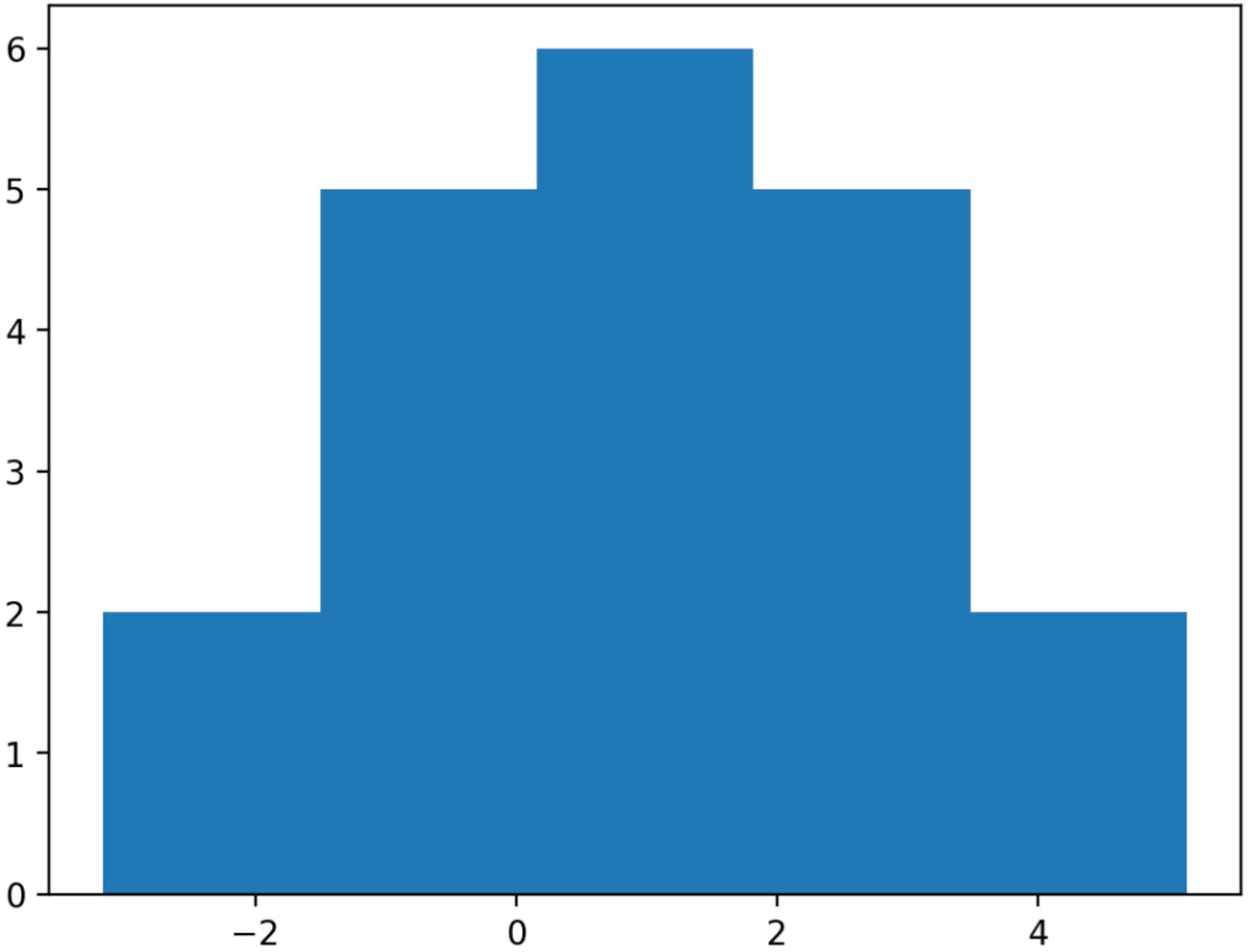
hist(): creates a histogram of the random numbers with 15 bins.

Fig: object is stores the histogram result

ax: object is used to manipulate the plot

st.pyplot() from Streamlit is used to display the histogram in the Streamlit app.

Plot with pyplot



Line and Bar Chart

```
import pandas as pd

df= pd.DataFrame(np.random.randn(10,
2),  columns=['x', 'y'])

st.line_chart(df)

st.bar_chart(df)
```

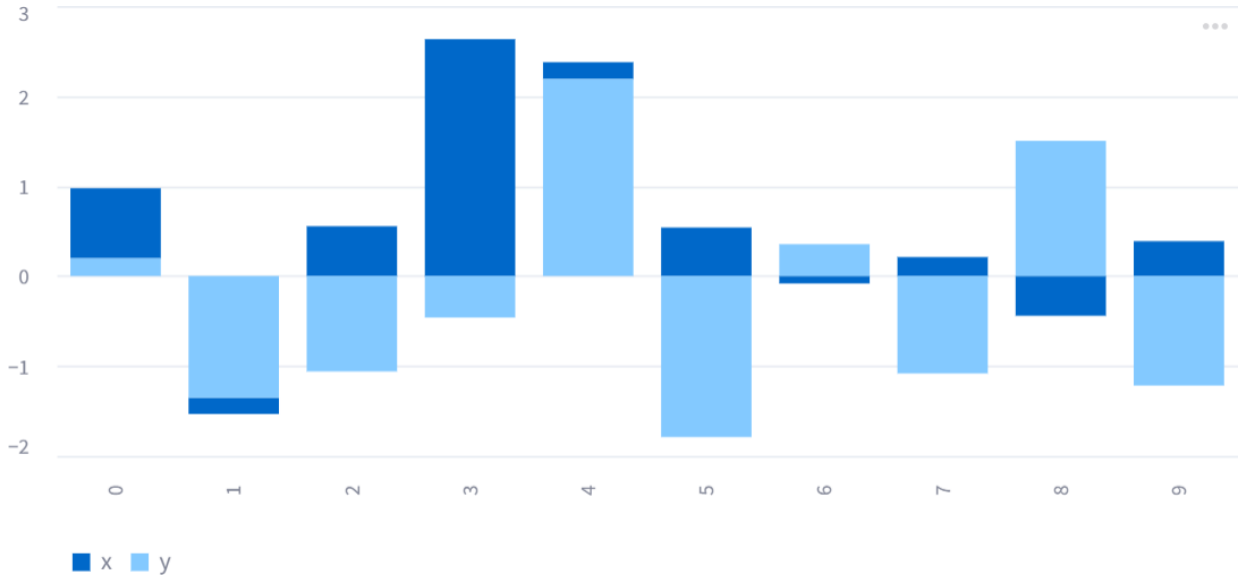
np.random.randn(): creates a Pandas DataFrame with 10 rows and 2 columns, where the values are randomly generated

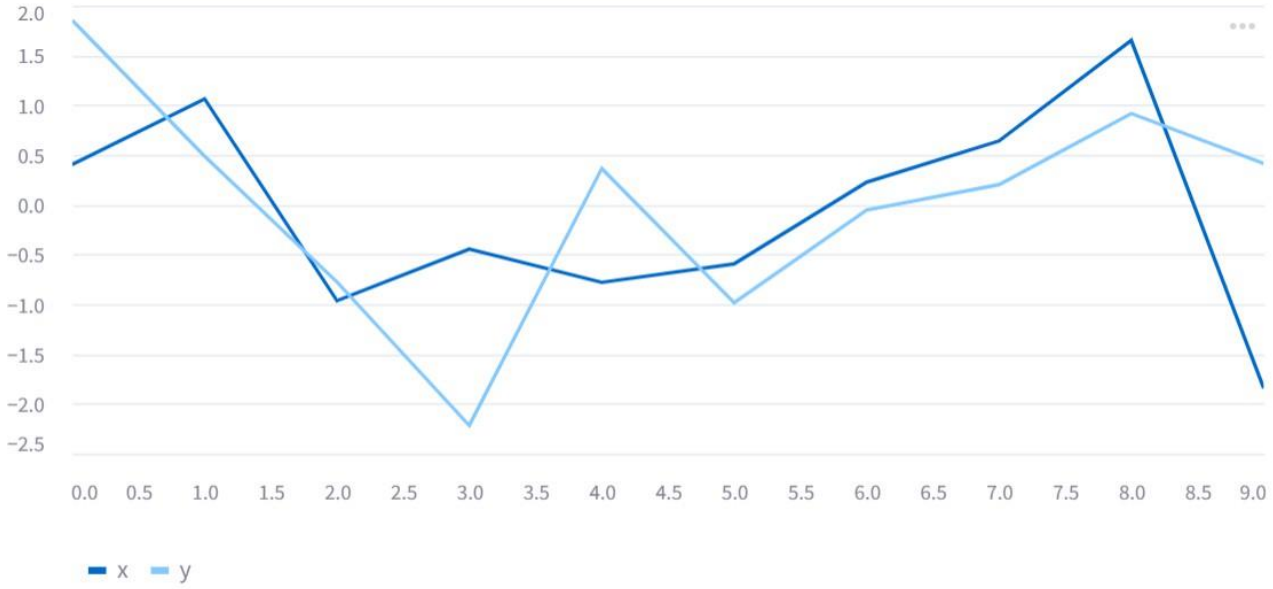
The columns are labeled 'x' and 'y'.

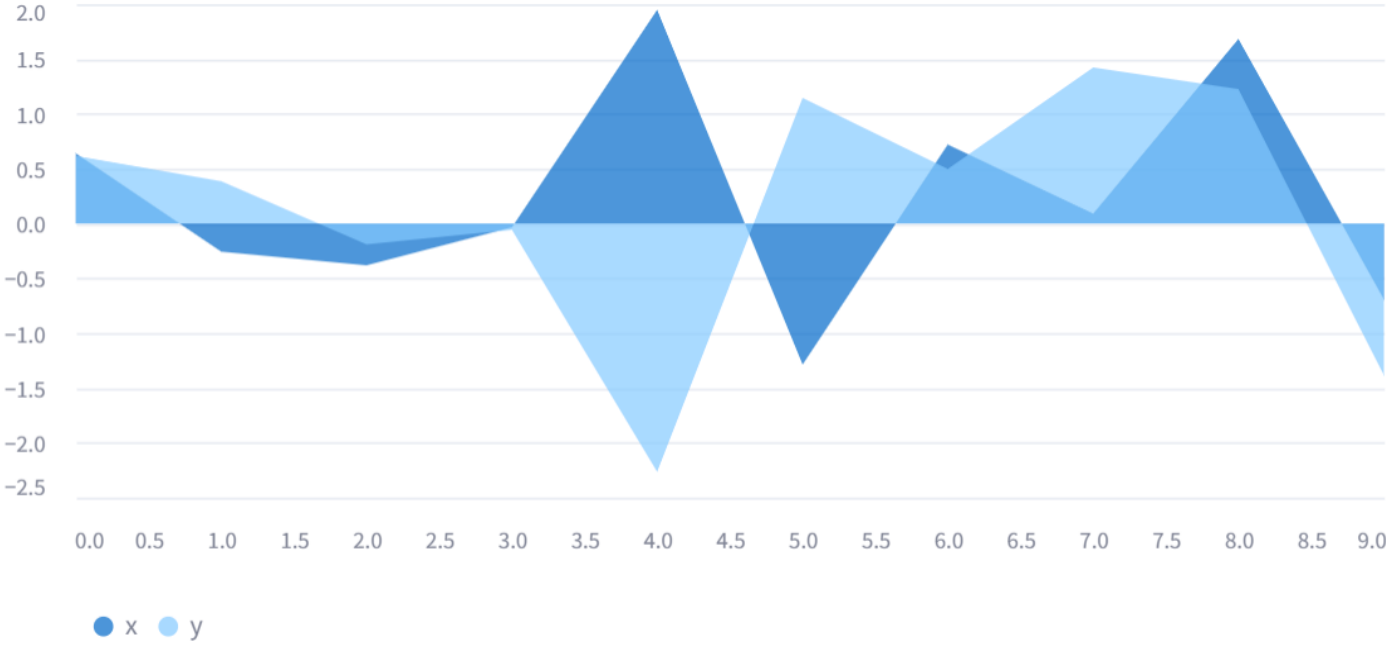
st.line_chart(): from the streamlit library to display a line chart

`st.bar_chart(df):` from the streamlit library to display a bar chart

Bar, Line, Area Chart



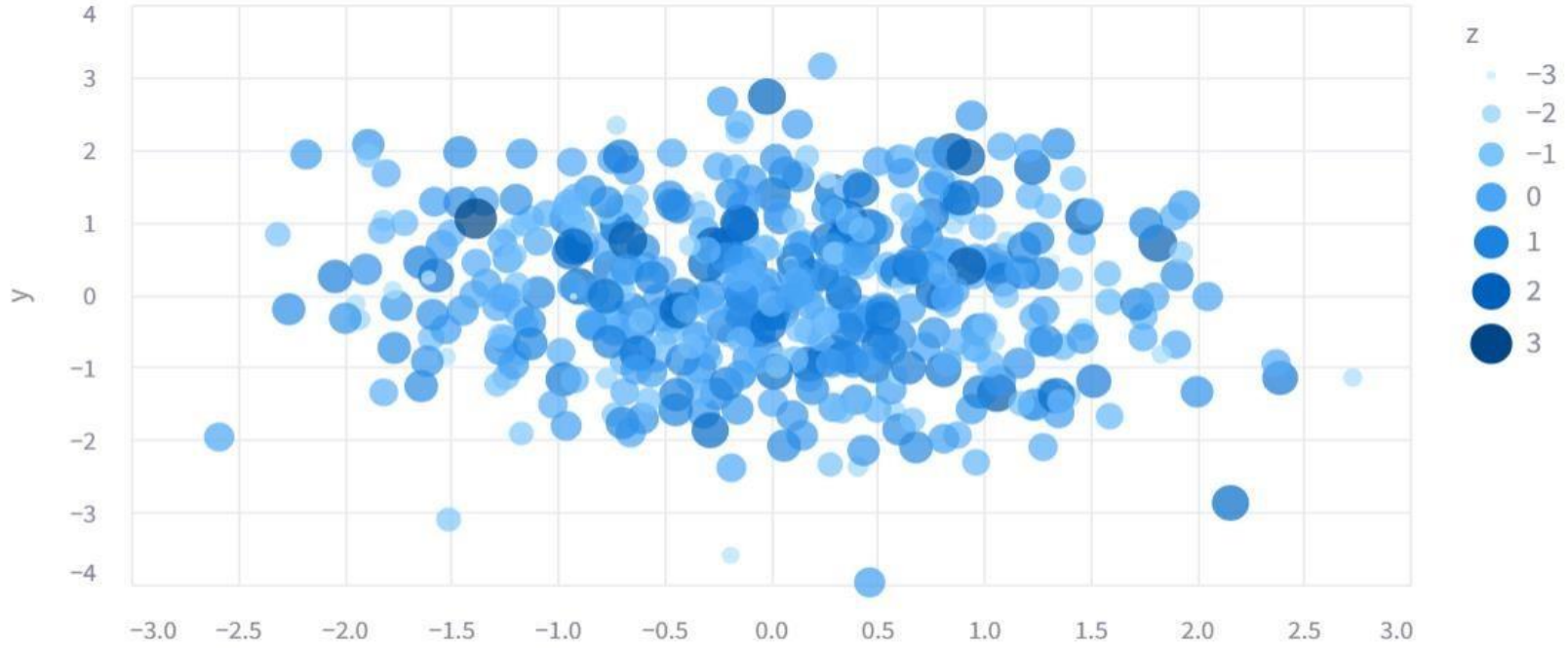




Plot with Altair

```
import altair as alt

df= pd.DataFrame(np.random.randn(500, 3),    columns=['x','y','z'])
c = alt.Chart(df).mark_circle().encode(    x='x', y='y' , size='z', color='z',
tooltip=['x', 'y', 'z'])
st.altair_chart(c, use_container_width=True)
```

Plot with Altair



Display maps with Streamlit

- ***st.map()***: This function is used to display maps in the app.

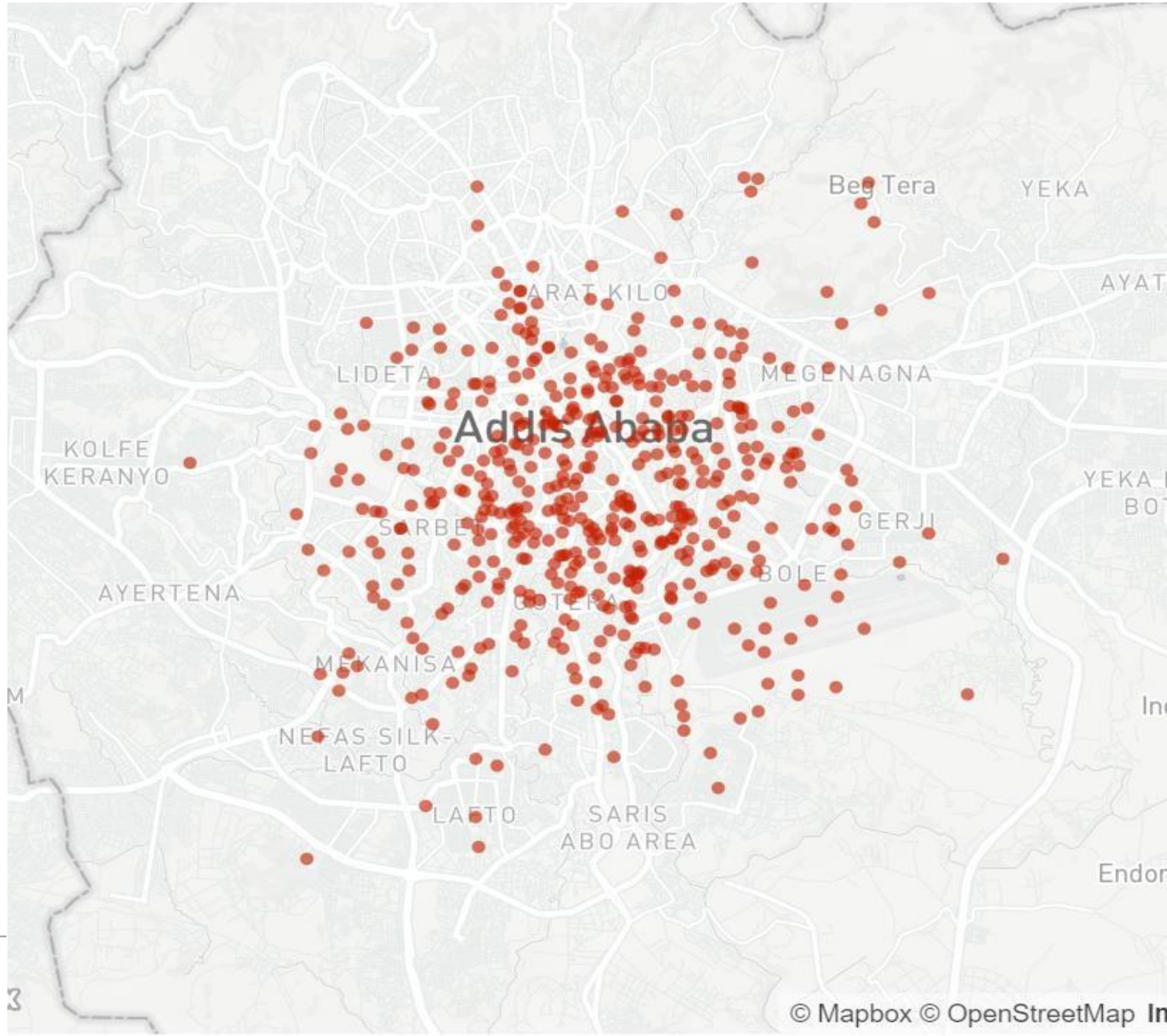
- Create a pandas DataFrame with 500 rows and 2 columns, where the values are randomly generated.

- The values are then divided by [50, 50] and added to the coordinates [9.00, 38.763] to create a set of latitude and longitude coordinates.

- The columns are labeled 'lat' and 'lon'.

- **st.map** function from the streamlit library is used to display the coordinates on a map.

Display Maps with Streamlit

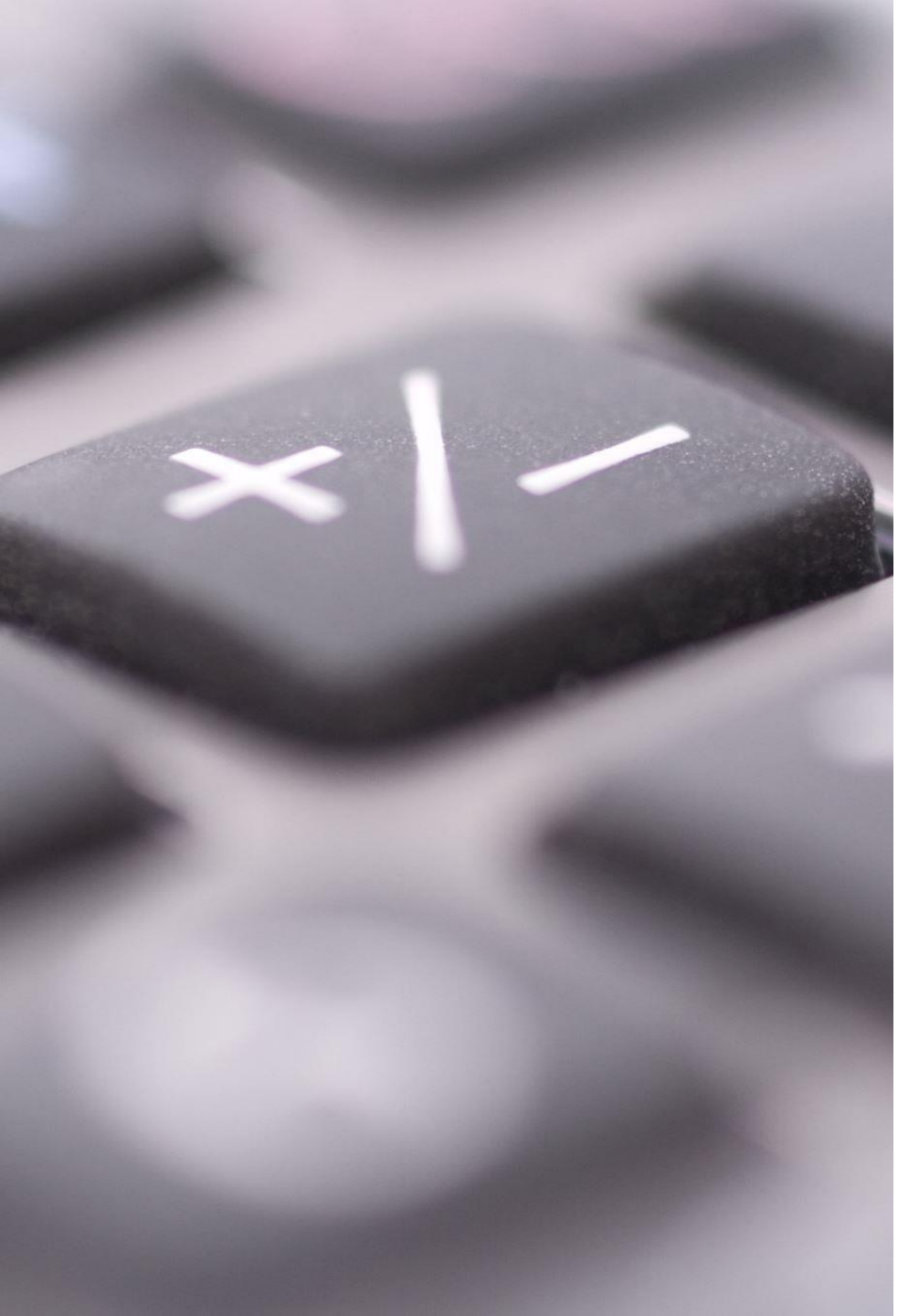


Change Themes

Example: BMI Calculator web app

- Let us recollect everything that we learn above and create a BMI Calculator web app.
- To calculate **BMI** Index the **weight is required in Kgs** and **height is in meters**

$$bmi = weight/height^2$$



BMI Calculator web app

- Import The Streamlit Library
- Give Title To The App
- Take Weight Input
- Take Height Input
- Calculate The BMI
- Print The BMI INDEX
- Interpret BMI Index

Example 1: BMI Calculator web app

```python
# Import the streamlit library
import streamlit as st

# Give a title to the app
#st.title('Welcome to BMI Calculator')

# blue, green, orange, red, violet, gray/grey, rainbow.
st.title(':blue[Welcome  to BMI Calculator]  🎛 ')

# TAKE WEIGHT INPUT in kgs
weight = st.number_input("Enter your weight (in kgs)")

# TAKE HEIGHT INPUT
# radio button to choose height format
status = st.radio('Select your height format ', ('cms',
'meters', 'feet'))

# take height input in centimeters

if(status == 'cms'):

    height = st.number_input('Centimeters')

    try:
        bmi = weight / ((height/100)**2)
    except:
        st.text("Enter some value of height")
```

```python
elif(status == 'meters'):
    # take height input in meters
    height = st.number_input('Meters')

    try:
        bmi = weight / (height ** 2)
    except:
        st.text("Enter some value of height")

else:
    # take height input in feet
    height = st.number_input('Feet')

    # 1 meter = 3.28 feet
    try:
        bmi = weight / (((height/3.28))**2)

    except:
        st.text("Enter some value of height")

# check if the button is pressed or not
if(st.button('Calculate BMI')):

    # print the BMI INDEX
    st.text("Your BMI Index is {}.".format(bmi))
```

BMI Calculator web app

```
if(st.button('Interprate BMI Result')):
    if(bmi < 16):
        st.error("You are Extremely Underweight")
    elif(bmi >= 16 and bmi < 18.5):
        st.warning("You are Underweight")
    elif(bmi >= 18.5 and bmi < 25):
        st.success("Healthy")
    elif(bmi >= 25 and bmi < 30):
        st.warning("Overweight")
    elif(bmi >= 30):
        st.error("Extremely Overweight")
```

BMI Calculator web app

```
if(st.button('Interprate BMI Result')):
    if(bmi < 16):
        st.error("You are Extremely Underweight")
    elif(bmi >= 16 and bmi < 18.5):
        st.warning("You are Underweight")
    elif(bmi >= 18.5 and bmi < 25):
        st.success("Healthy")
    elif(bmi >= 25 and bmi < 30):
        st.warning("Overweight")
    elif(bmi >= 30):
        st.error("Extremely Overweight")
```

How to deploy a Streamlit App

Deploying an application is the process of ***copying***, ***configuring***, and ***enabling*** a specific application to a specific base URL.

Deployment is the mechanism through which applications are delivered from developers to users.

Once the deployment process has finished, the application becomes publicly accessible on the base URL.

The server carries out this two-step process by first **staging** the application, and then **activating** it after successful staging.

Create a **GitHub Account:** https://github.com/

Create a **new repository** on your GitHub where you need to put your app code and dependencies.

How to deploy a Streamlit App

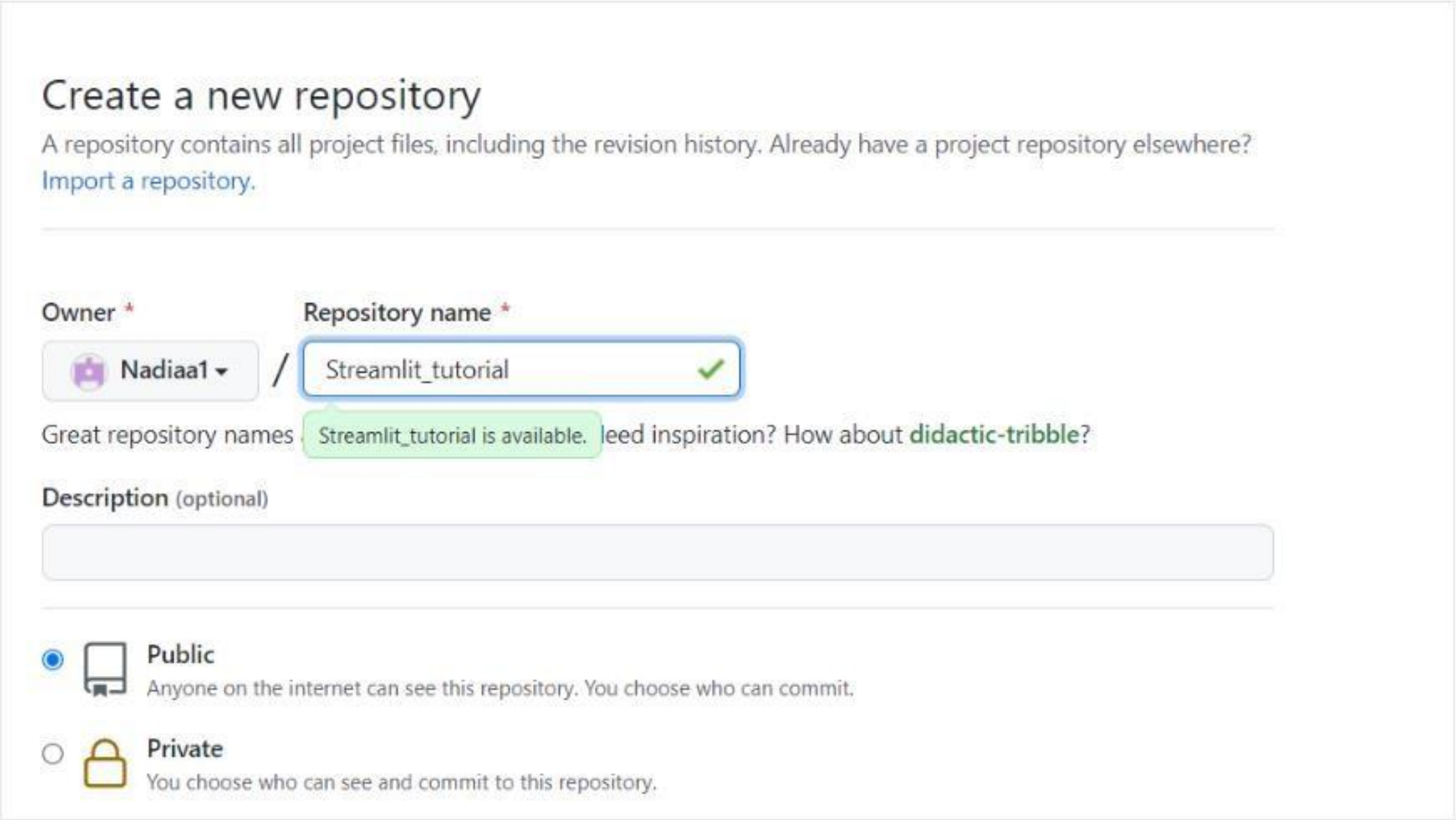
Upload your codes and data to the newly created repository

Create a new file named requirements where you have to put the libraries you used in your app.

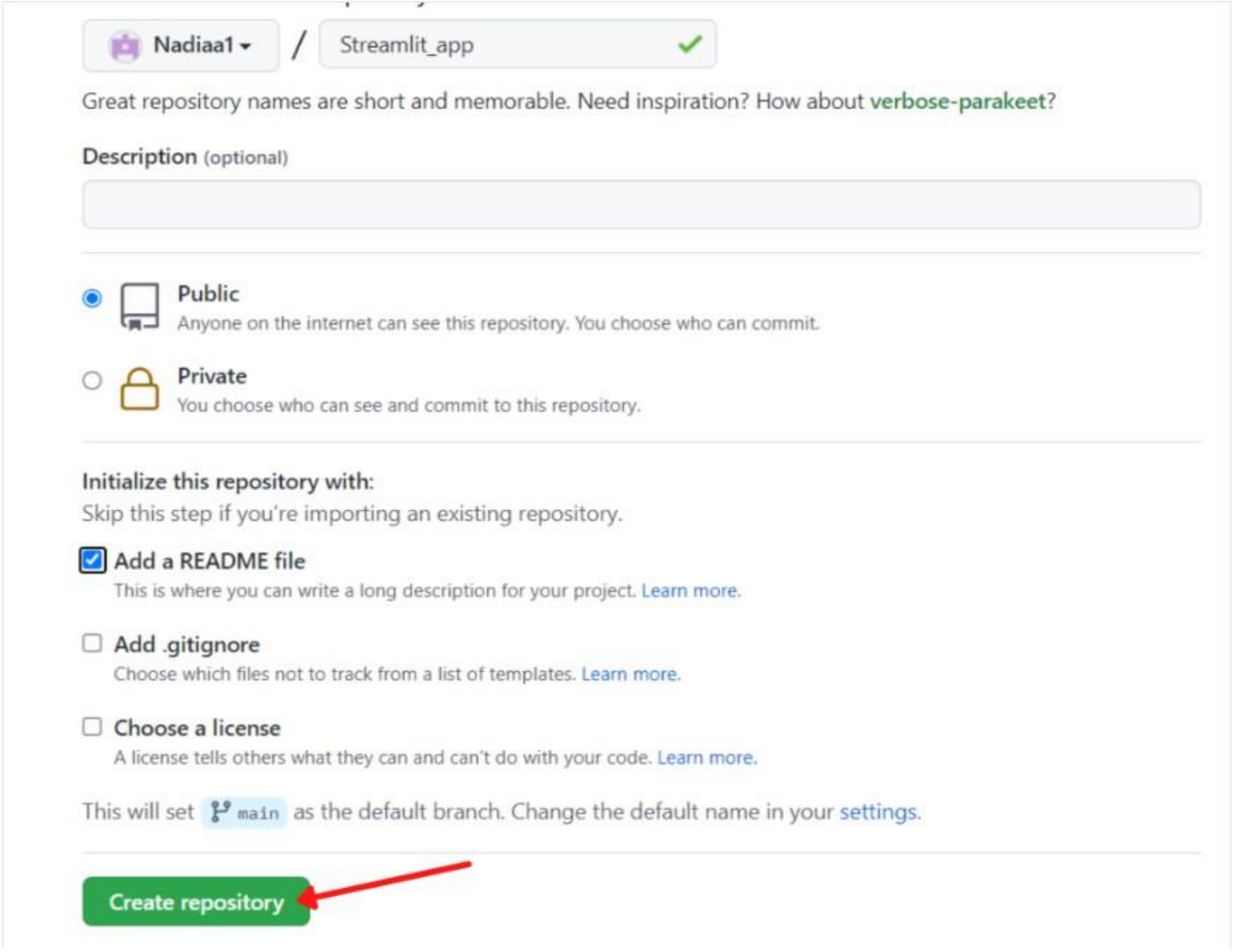
Go to this website and link your GItHub account with the streamlit cloud: https://share.streamlit.io/

On the streamlit cloud: New App, then Deploy

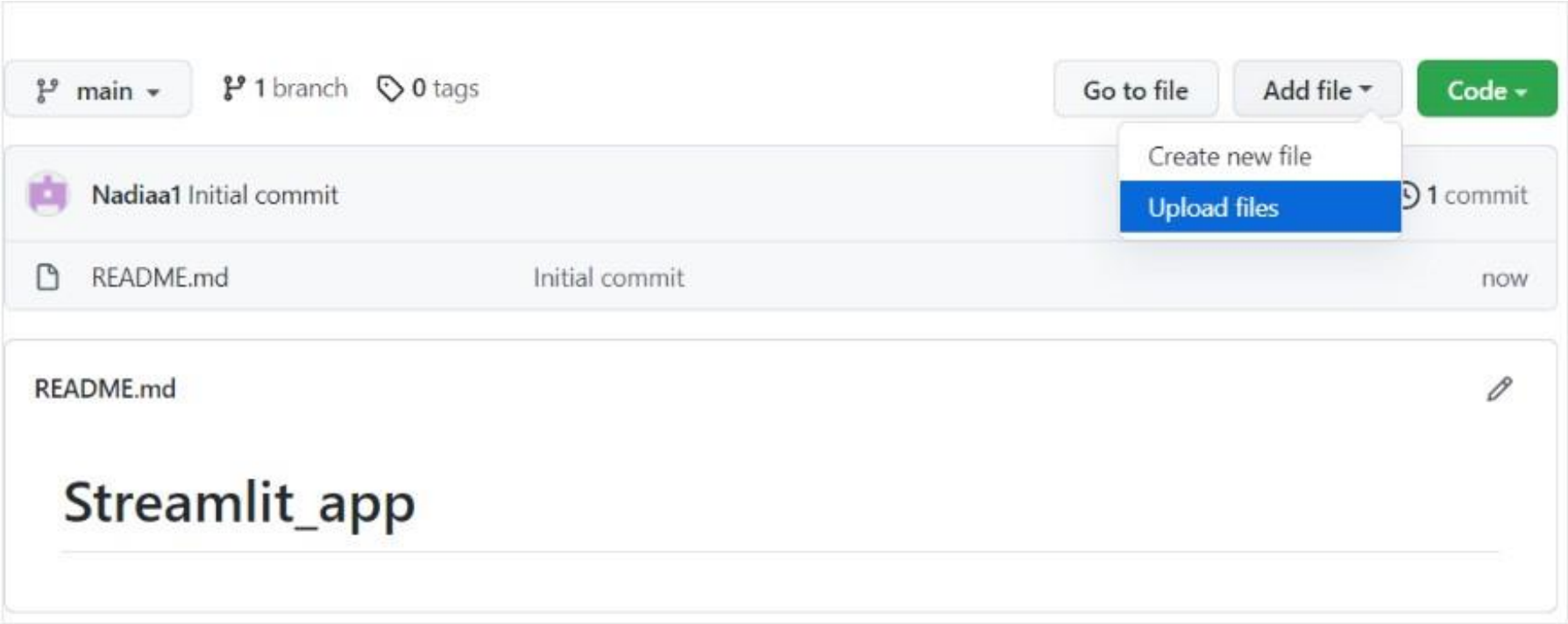
Create New GitHub Repository

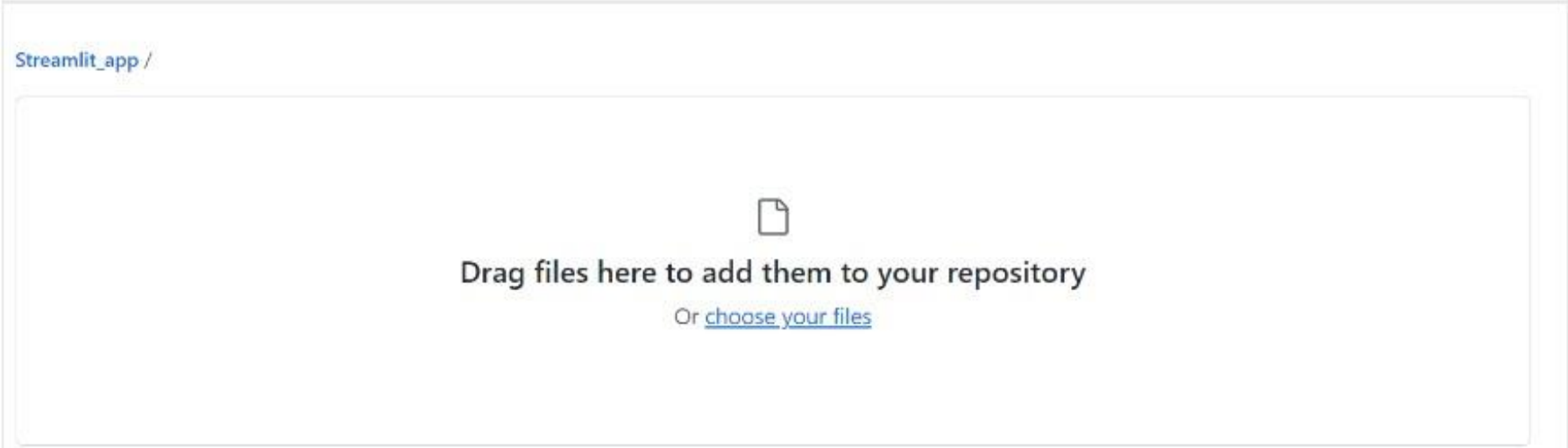


Create New GitHub Repository

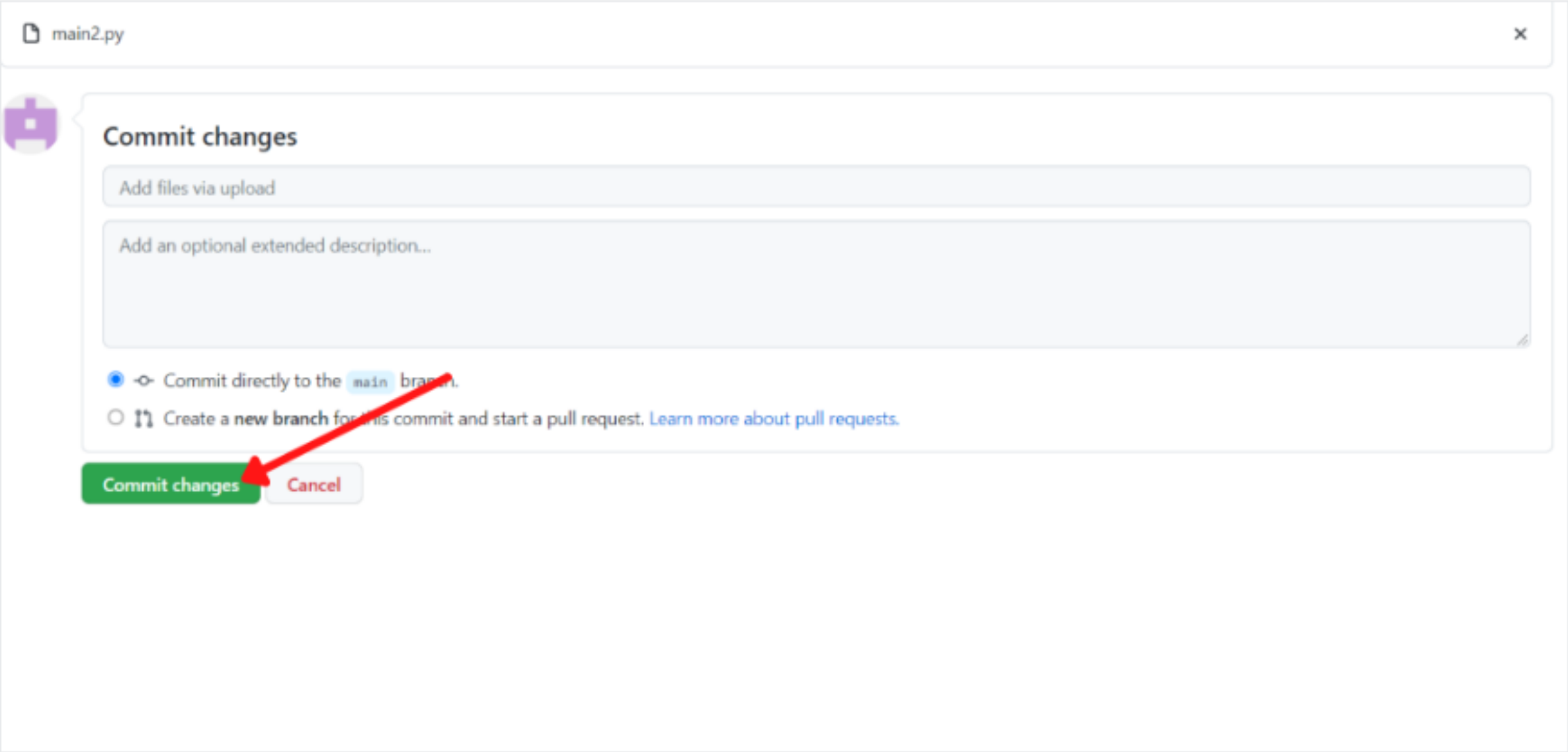


Upload files to GitHub Repository





Commit Chages



Create requirements file

- A ***requirement file*** is a type of file that usually stores information about all the libraries, modules, and packages specific to the project used while developing a particular project.

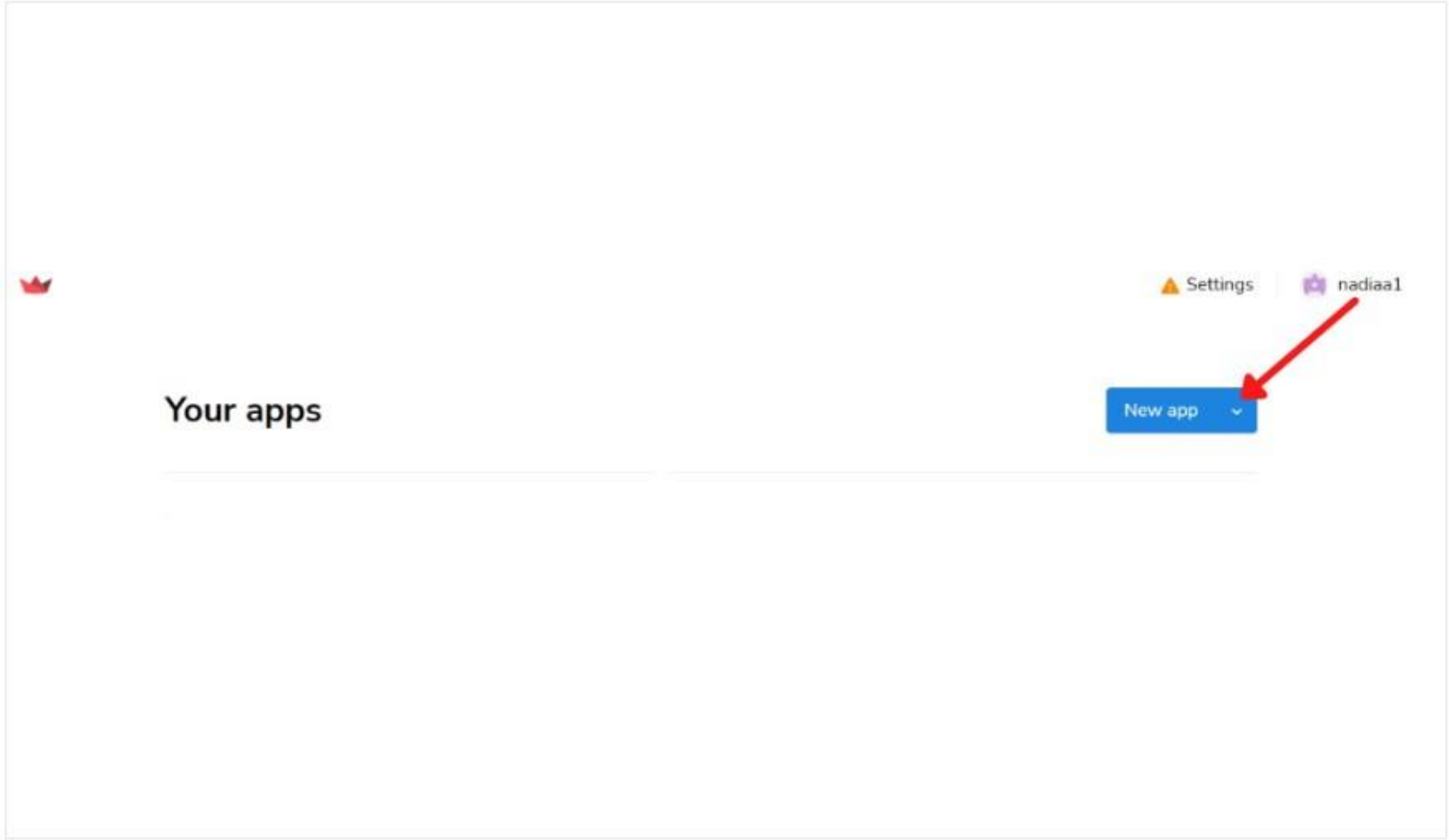
  - pip install freeze

  - pip freeze >requirements.txt (not recommended)
  - pip freeze |grep -i panda >>requirements.txt (better) [works only in Lunix]

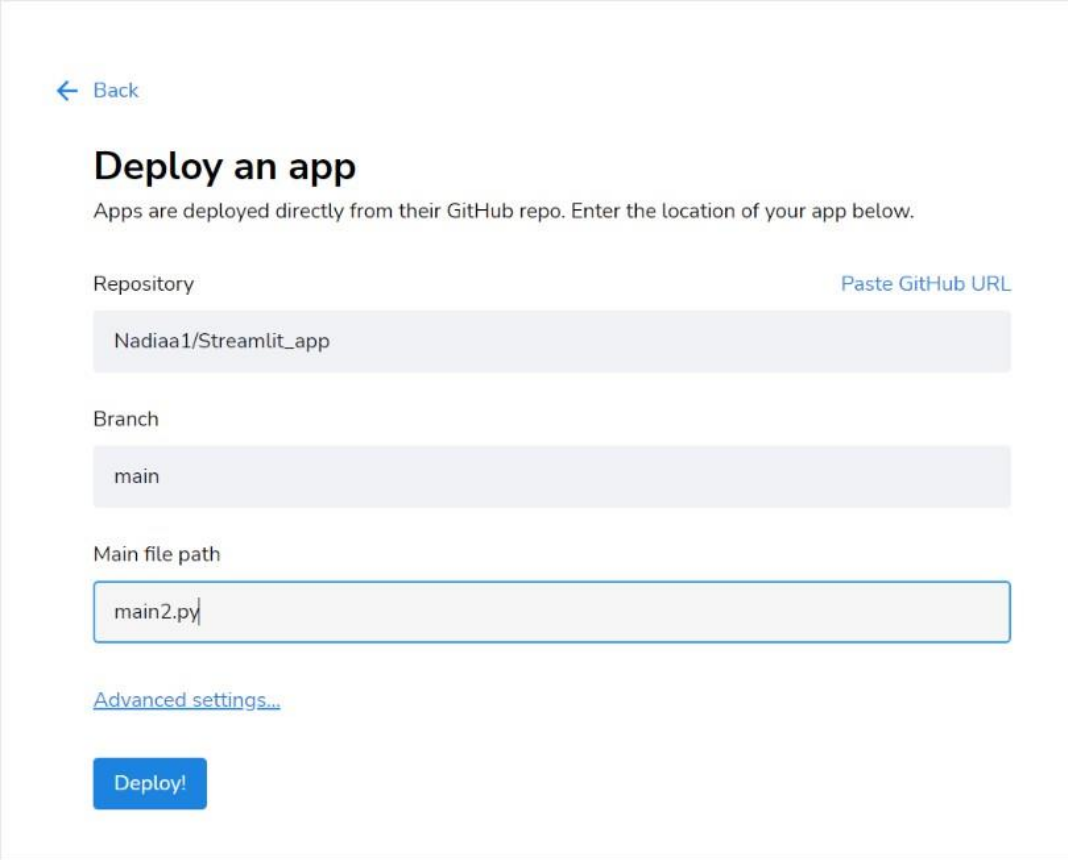
  - pip install pipreqs

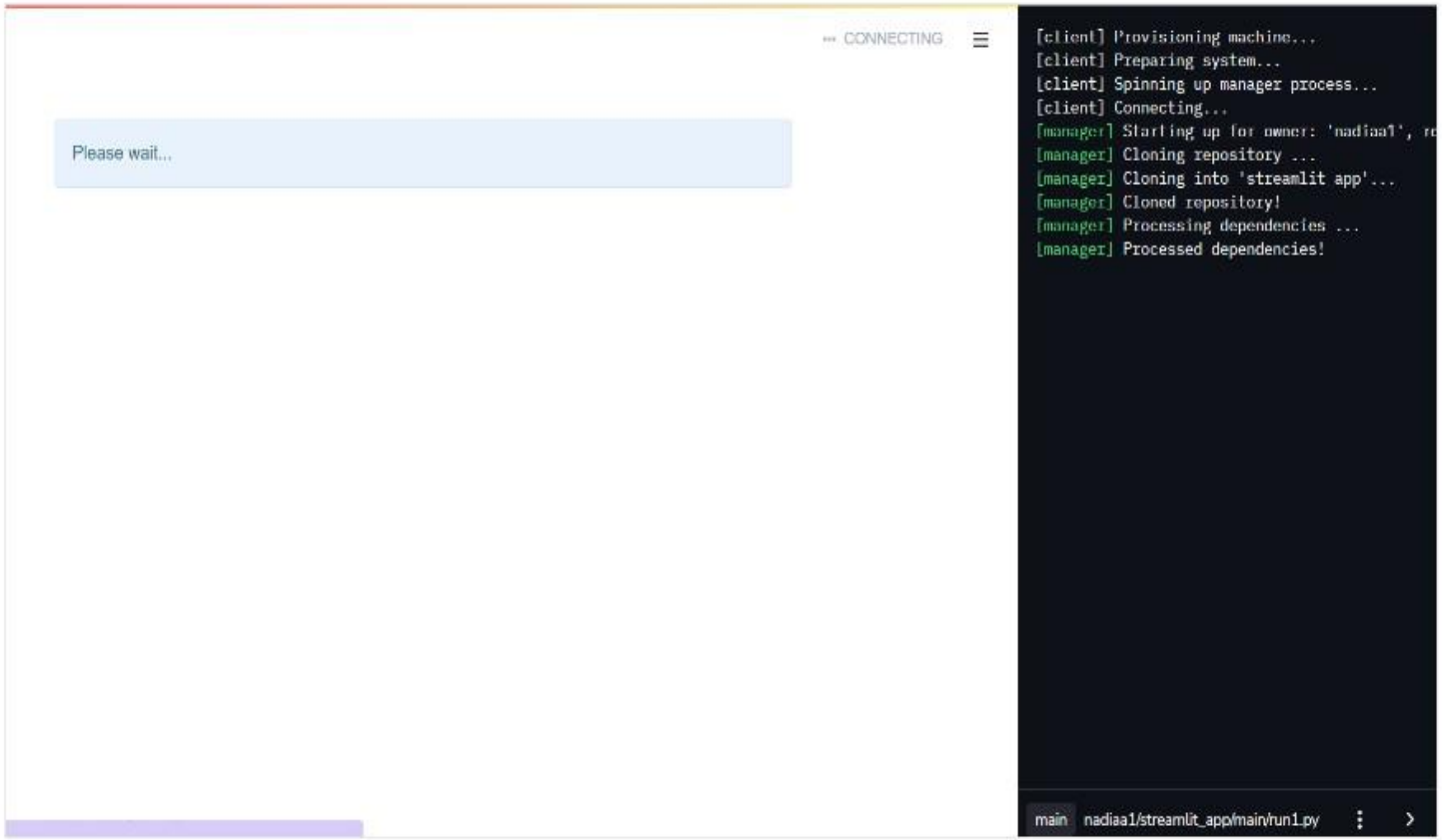
  - pipreqs
  - pipreqs [Path]

Deploying your app



Deploying your app





Links

- Github repository for the ML model:https://github.com/abelataguta/Transportation_project_training

  ML Model deployed on Streamlit Cloud: /

- Email: abelataguta@gmail.com

Thank You

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Happy web app deployment